Physiology & Behavior 86 (2005) 427 – 433

PHYSIOLOGY & BEHAVIOR

Circadian activity rhythms in the spiny mouse, *Acomys cahirinus*

E.T. Weber *, V.M. Hohn

Department of Biology, Rider University, 2083 Lawrenceville Road, Lawrenceville, NJ 08648, USA

Received 5 November 2004; accepted 6 July 2005

Abstract

Circadian locomotor rhythms were examined in adult common spiny mice, *Acomys cahirinus*. Spiny mice demonstrated nocturnal activity, with onset of activity coinciding promptly with onset of darkness. Re-entrainment to 6-h delays of the light–dark cycle was accomplished faster than to 6-h advances. Access to running wheels yielded significant changes in period and duration of daily activity. Novelty-induced wheel running had no effect on phase of activity rhythms. Circadian responses to light at various times of the circadian cycle were temporally similar to those observed in other nocturnal rodent species. No gender differences were observed in any of the parameters measured.

Keywords: Circadian rhythms; Locomotor activity; Spiny mouse; Suprachiasmatic nuclei; Entrainment; Wheel-running activity; *Acomys cahirinus*

1. Introduction

Comparative studies describing unique phenomena associated with the circadian system of particular species provide opportunities for elucidating the physiological basis for general properties of circadian systems in mammals. For example, alternation of diurnal-nocturnal activity preferences in *Octodon degus* [3], *Arvicanthis niloticus* [4], and *Acomys russatus* [5,6] in response to variable social or environmental conditions potentially provide insight into the mechanisms responsible for the seemingly inflexible manifestation of nocturnal versus diurnal phenotypes observed in most other species. In Syrian hamsters (*Mesocricetus auratus*), tremendously potentiated phase shifts are often observed after novelty-associated wheel-running activity [7] or in response to pharmacological manipulations [8–10]. These and similar observations may provide insight into mechanisms underlying disruptions experienced as jetlag, delayed onset sleep disorders, etc., that affect some but not all humans subjected to circadian system challenges.

The common spiny mouse, *Acomys cahirinus*, is indigenous to arid regions from western Asia to southern Africa. The physiological and behavioral adaptations to the challenges found in these conditions make spiny mice an interesting subject of comparative studies in strategies of water conservation, temperature regulation, olfactory and gustatory identification and preference development, social preferences, aggression, diabetes, ingestive behavior, and exploratory behavior [11]. With regards to circadian rhythms research, prenatal maternal influences on entrainment have been demonstrated using the precocious offspring of spiny mice which express their own directly entrainable circadian systems on the first day of birth [12–14]. This attribute of spiny mice provides a rare opportunity for studying the development of the mammalian circadian system.

A careful examination of the basic properties of circadian biology has not been described in spiny mice. The aim of the experiments in this report was to describe the circadian locomotor activity of adult common spiny mice, *A. cahirinus*. We describe here the influences of manipulations of lighting conditions and of effects of running wheels on the general features of circadian activity rhythms in adult male and female spiny mice—periodicity, duration of activity, intensity of activity, phase angle of entrainment, rate of re-entrainment.

* Corresponding author. Tel.: +1 609 896 5028; fax: +1 609 895 5782.
E-mail address: tweber@rider.edu (E.T. Weber).

0031-9384/$ - see front matter © 2005 Elsevier Inc. All rights reserved.
doi:10.1016/j.physbeh.2005.07.023

2. Materials and methods

2.1. Animal housing

Adult male and female spiny mice donated from the inbred breeding colony of Dr. Richard Deni at Rider University were the subjects of these experiments. After weaning, spiny mice were group-housed with same-sex siblings in the main colony, 3–6 mice per cage, under a 12:12-h light/dark cycle (LD12:12) of 200–400 lx fluorescent white room illumination. Food (Purina rat chow #5012) and water were provided ad libitum at all times. Cages were changed weekly, using Alpha-Dri bedding material (Shepherd Specialty Papers, Waterhouse, TN). All procedures were approved by the Rider University Institutional Animal Care and Use Committee in accordance with the regulations of the Animal Welfare Act.

2.2. Activity rhythm measurements

To monitor activity, spiny mice were housed individually in a dedicated experiment room in cages equipped with infrared motion detectors (Slimline 12 V detectors, SmartHomePro.com). In some experiments, mice were given access to running wheels equipped with hermetic switches activated by a magnet attached to a spoke on the wheel. For both activity recording techniques, daily activity of the mice was constantly monitored via Clocklab data collection software (Actimetrics Inc., Evanston, IL) running continuously on a Dell computer. Circadian data analyses were performed with Clocklab data analysis software. Rhythms were monitored in both male and female spiny mice, as gender differences in other aspects of spiny mouse physiology and behavior (metabolism, exploratory behaviors) have been documented [15,16].

Daily activity onsets were defined as a (minimum) 1-h bout of uninterrupted activity following 3 h of inactivity (less than 5 counts/5 min bin), while offsets were determined as 2-h bout of inactivity following (minimum) 1 h of activity. Duration of daily activity (alpha) was calculated as the difference in time between onsets and offsets. The period of any particular portion of an animal's rhythm was determined using the Clocklab analysis program by fitting a regression line through onsets of activity for 5–7 days, and confirmed by periodogram analysis.

Entrainment, rates of re-entrainment, properties of free-running rhythms, and light responsiveness (in that order) were examined in Experimental Group 1, which consisted of 16 spiny mice ($n=8$ males and $n=8$ females, 3–4 month old adults).

2.3. Entrainment

To examine rates of re-entrainment, activity of Experimental Group 1 mice monitored by motion detectors was assessed following a 6-h advance of the room lighting cycle. The number of days required for stable entrainment to the new light–dark cycle was determined by counting days following treatment until the onset of activity for each day occurred within the average phase angle of entrainment for that mouse and persisted for at least three successive days. After all animals were re-entrained for 10–20 days, mice were exposed to a 6-h delay in the light–dark cycle. Comparisons of rates for males and females were assessed for each treatment using Student's *t*-tests.

2.4. Free-running period

Following the phase shifts described above, the 16 spiny mice in Experimental Group 1 (being monitored by motion detectors) remained entrained to 12:12LD for 30 days before being released into constant darkness (DD). To determine whether wheel-running activity has effects on expression of activity rhythms, half of the spiny mice used in the experiments described above ($n=4$ males and $n=4$ females) were given access to exercise wheels monitored concurrently with general activity measurements via motion detectors, while the remaining 8 mice were given immobilized wheels. Wheels were introduced to the cage during the dark portion of the LD cycle, to minimize potential "novelty" effects associated with wheel introduction (see Novel Wheel section). Changes in duration of daily activity (alpha) and in free-running period for each group were compared by two-way repeated measures ANOVA (alpha or period × gender × wheel), followed by Tukey analysis when appropriate. Following this phase of behavioral monitoring, wheels were removed from all cages.

2.5. Light phase responsiveness

To determine the responsiveness of the circadian system to light exposure at various times of the circadian cycle, spiny mice in Experimental Group 1 were maintained under constant darkness for 3 weeks following removal of wheels from their cages. The mice were concurrently exposed in their home cages to room lighting (250–400 lx fluorescent white at the level of each mouse) for 30 min. The circadian time (CT) of light exposure for each mouse was then determined post hoc by comparing the timing of light exposure to each animal's activity onsets prior to the treatment. ("Circadian time" refers to timepoints relative to a circadian phase marker, with 1/24 of the length of a circadian cycle representing a "circadian hour". In these experiments, circadian time of light exposure was determined relative to daily onset of activity, conventionally defined as circadian time 12 (CT12).) During the light exposures, animals were monitored every 2–3 min and gently prodded if no motion or whisker movements were observed, to prevent animals from sleeping during the treatment. Following light exposure and return to DD, activity was monitored until stable phase had been re-established for a minimum of 10 days. Phase shifts in response to treatments were calculated as the difference in predicted onset of activity prior to treatment, and deduced onset of activity based on subsequent phase extrapolated back to the day of treatment, as described previously [17]. Once stable periods had been re-established for all spiny mice, animals were re-treated, so that over the course of this phase of the experiment, mice were re-exposed to light a total of five times, at approximately 18-day intervals.

2.6. Novel wheel effects

To determine whether spiny mice also demonstrate novel wheel-induced phase shifts, an additional group of adult male and female spiny mice (Experimental Group 2, $n=3$ each, age 3–4 months) housed under constant darkness were given a running wheel at CT 6. Initially, mice were individually housed in the experiment room under 12:12 light–dark cycles and monitored by motion detectors. After stable entrainment was established, mice were released to constant darkness for 10 days. In the middle of their subjective daytime (approximately CT6), each mouse was given a functional running wheel which remained in its cage for the duration of the experiment. Shifts in phase following introduction of the wheel were calculated as described above.

3. Results

3.1. Entrainment

Spiny mice in our experiments all displayed nocturnal activity patterns. Subjects entrained to a 12:12 light–dark cycle, with onset of activity varying within 11±6 min around onset of darkness each day (Fig. 1, Table 1). No gender differences in the phase angle of entrainment, such as the estrous cycle-related "scalloping" seen in female hamster rhythms [18], were apparent. For 2 of 22 animals studied, bouts of activity were observed to precede onset of darkness (average 12±2 min) for 8–10 days following placement in individual monitoring cages (Fig. 1, left panel), but because the bouts of activity were not continuous for at least 1 h, they were not considered to be the animals' daily onsets of activity. Initially, activity persisted with relatively constant intensity throughout the dark portion of the light–dark cycle and diminished with onset of light, although 5 of 22 mice in the study ($n=16$ from Experimental Group 1 and $n=6$ from Experimental Group 2) displayed activity concentrated either early or late in their daily activity bouts at some time during the course of activity monitoring.

Table 1
Gender comparison of properties of circadian rhythms in spiny mice

Variable	Male (*n*)	Female (*n*)	*p*
Phase angle of entrainment (min)	10.2±0.8 (11)	11.0±0.6 (11)	0.2
Duration of daily activity (h)			
12:12 LD	11.5±0.1 (8)	11.7±0.3 (8)	0.5
1 week in DD	11.4±0.2 (7)	11.6±0.1 (8)	0.5
6 weeks in DD	13.35±0.4 (8)	14.1±0.5 (8)	0.3
Rates of re-entrainment			
6-h advance	10.1±1.8 days (8)	11.8±1.1 days (8)	0.6
6-h delay	5.0±0.8 days (8)	4.6±0.9 days (8)	0.5
Free-running period (h)			
Expt Group 1–after 1 week	24.03±0.03 (7)	24.06±0.02 (8)	0.4
–after 6 weeks	24.18±0.06 (7)	24.16±0.05 (8)	0.4
Expt Group 2–after 1 week	23.92±0.03 (3)	23.93±0.02 (3)	0.7

Spiny mice responded to a 6-h advance in the light–dark cycle by gradually advancing the onset of daily activity to correspond once again to dark onset. Re-entrainment was achieved at a relatively constant rate over approximately 10 days, with no differences ($p>0.05$) observed between males

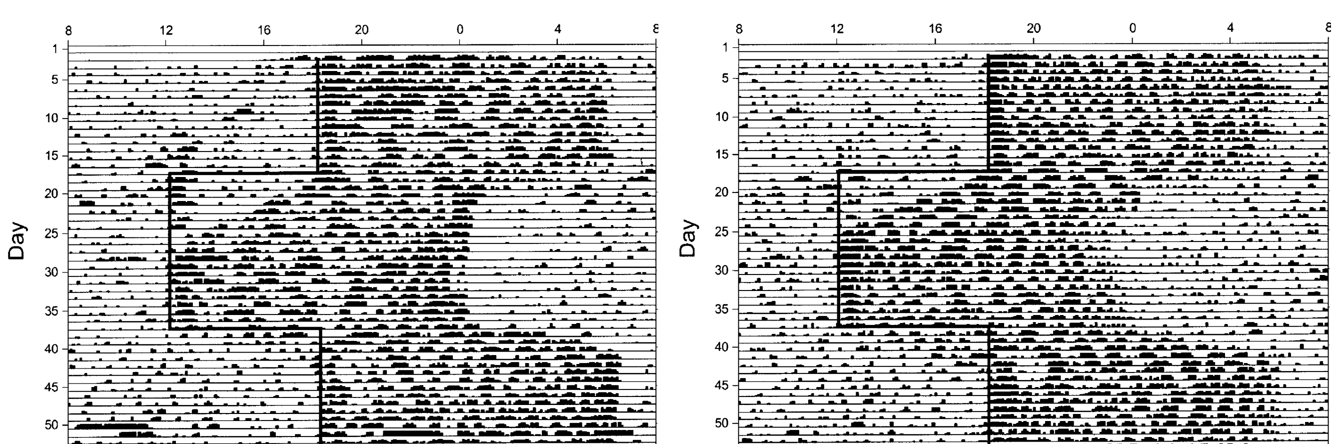

Fig. 1. Representative activity records collected from adult spiny mice (left panel, male; right panel, female) using infrared motion detectors show entrainment and re-entrainment to 6-h shifts in the light–dark cycle. Deflections from the baseline represent activity collected in consecutive 5-min samplings of a motion detector. Dark vertical lines superimposed on the actogram represent the onset of the dark portion of the 12:12 lighting cycle, which was advanced by six hours on day 17 of these records, then delayed by 6 h on day 37. No differences in rates of re-entrainment were observed for either phase advances or delays between males and females.

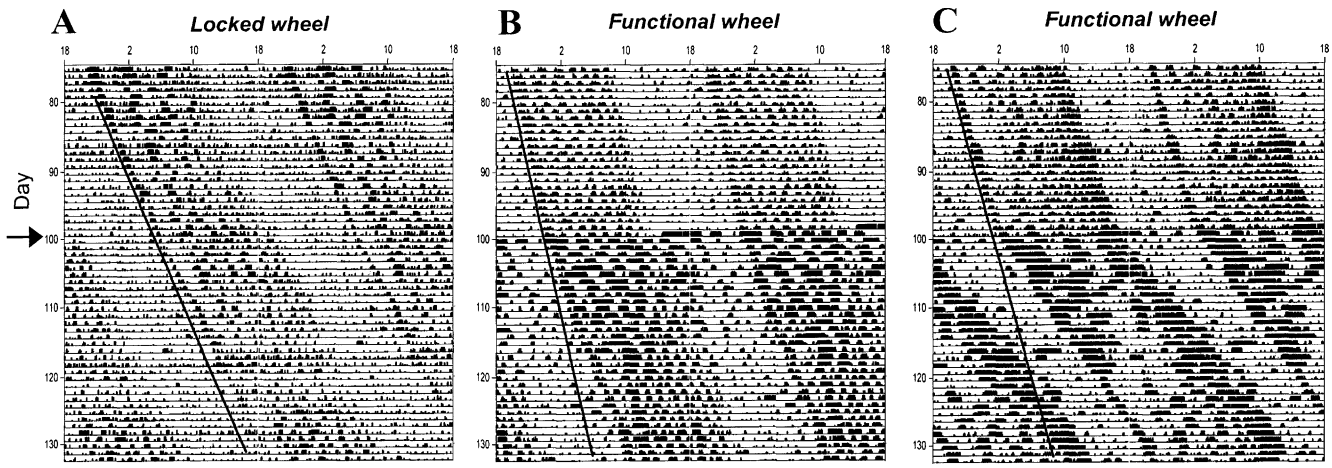

Fig. 2. Representative actograms demonstrate the gradual lengthening of the free-running period of activity rhythms of spiny mice housed in constant darkness. Data are double-plotted (both next to and below the previous day's activity) to enhance visualization of long-term changes. In these records, activity was monitored by motion detectors. On day 99, either (A) locked or (B, C) functional wheels were introduced (indicated by arrow at left). Free-running period increased dramatically in some mice immediately after introduction of functional wheels (panel C). Significant differences between groups of mice with and without functional wheels were seen at 5 weeks.

and females (Table 1). In 11 of the 16 mice in this experiment, short bouts of activity occurred briefly after dark onset during the first 4–5 days of the re-entrainment period (Fig. 1, left panel), most likely a masking effect of darkness. A 6-h delay in the light–dark cycle resulted in apparent re-entrainment of activity rhythms in approximately 5 days (Fig. 1), with no difference between males and females (Table 1).

3.2. Free-run in darkness

Both male and female spiny mice in Experimental Group 1 showed free-running periods of activity rhythms (monitored with motion detectors) which gradually increased from an average 24.05±0.03 h after 1 week in darkness to 24.17±0.06 h after 6 weeks in darkness (Fig. 2), with no differences evident between genders (Table 1). Addition of a functional running wheel increased free-running period almost immediately for 3 of the 7 spiny mice with functional wheels (Fig. 2C), with statistically greater increases in period for mice with functional wheels versus those with locked wheels by 3 months ($p<0.05$). Equipment malfunction precluded analysis of data from one of the mice. Duration of daily activity under constant darkness increased from 11.5±0.1 h after the first week in constant darkness to 13.8±0.3 h by the sixth week and was consistent among male and female mice (Table 1). Duration continued to increase for all mice over the following 6 weeks, with significantly greater increases in mice given functional running wheels (16.7±0.9 h/day) versus those given locked wheels (14.6±0.8 h/day) ($p<0.05$).

3.3. Light phase response curve

To establish a phase response curve for light, spiny mice housed in constant darkness for 8 weeks (without wheels) were exposed to 30 min of white room light at approximate 2–3 week intervals. These mice demonstrated shifts in onset of activity rhythms similar to those observed in other rodent species following similar light exposure (Fig. 3). In general, light exposure during early subjective night yielded phase delays, light exposure during late subjective night yielded

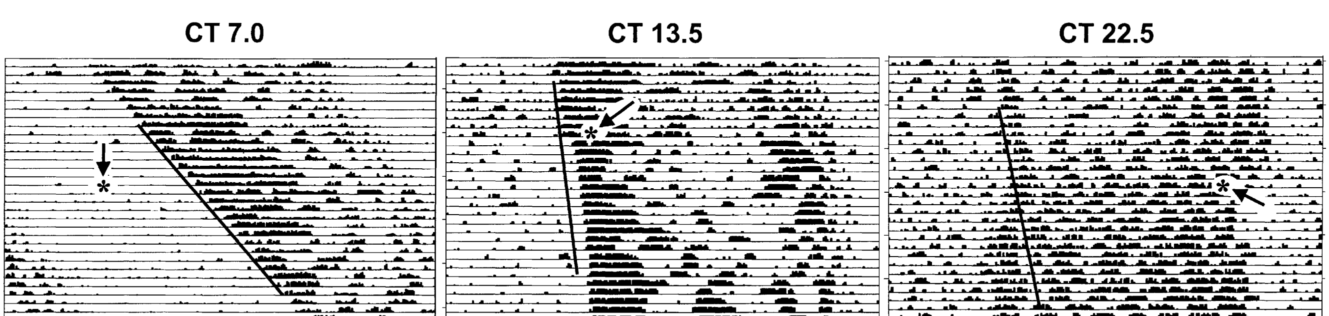

Fig. 3. Representative actograms show the effects of light exposure during mid-subjective day (CT7.0), early subjective night (CT13.5), and late subjective night (CT22.5), on free-running period of activity rhythms in spiny mice. Activity was monitored by motion detectors. Lines superimposed on the actograms represent predicted phase of activity rhythms had no treatment been administered, based on regression of activity onsets for 7 days prior to exposure to light. Time of light exposure is indicated by asterisks with arrows.

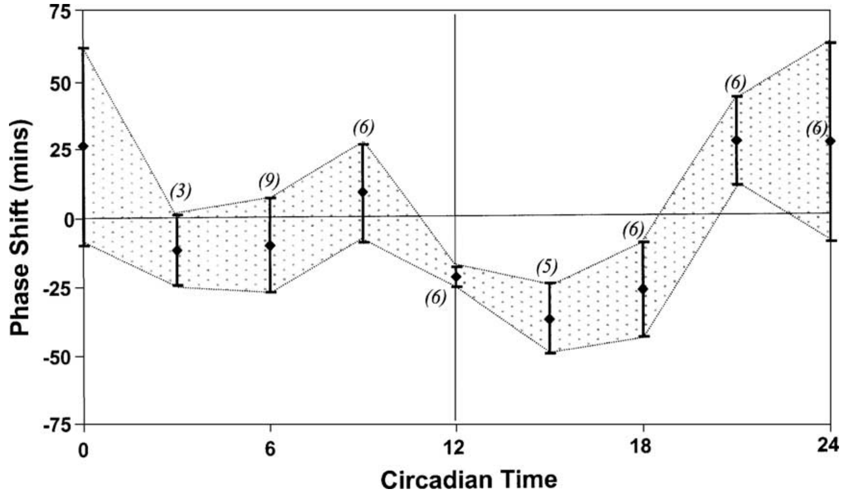

Fig. 4. A light phase response curve conveys the average shifts of onset of activity in response to room light exposure, based on the time of exposure in each mouse's circadian cycle (determined post hoc following each treatment). Timepoints are pooled 3-h averages ± SEM. The number of individual shifts per timepoint is indicated in parentheses.

phase advances, and light exposure during subjective day yielded small or no phase shifts (Fig. 4). The average magnitude of phase advances following single exposures to light under constant conditions was consistent with the average daily advances of spiny mice re-entraining to 6-h advances of the light–dark cycles. The phase delays following single acute light exposure under DD conditions were noticeably smaller than apparent daily phase delays observed during re-entrainment following a 6-h delay in the light–dark cycle.

3.4. Novel wheel introduction

Spiny mice free-running in constant darkness that were given access to functional running wheels in mid-subjective day (CT 6) responded with initial bouts of activity ranging from 299 to 3249 revolutions lasting 1.5–4.0 h in duration (Fig. 5), averaging 590 ± 125 revolutions/h (mean ± SEM). The amount of wheel-running activity was not related to gender. No changes in phase of free-running activity rhythms were observed in any of the six spiny mice over the following 2 weeks (Fig. 5).

4. Discussion

Based on the evaluation of activity of 22 male and female spiny mice from our colony, spiny mice are nocturnal, with activity consistently beginning promptly following the onset of darkness. Occasional anticipatory activity was observed, but was neither gender-specific nor regular in its occurrence. Spiny mice are avid runners on running wheels, which significantly increased the duration of daily activity and the period of free-running rhythms in constant darkness over 5–

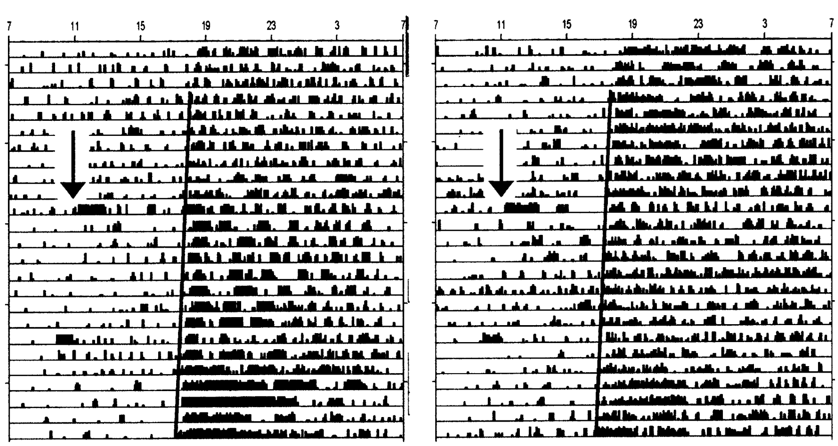

Fig. 5. Novel wheel running activity had no effect on free-running period of activity rhythms of spiny mice. Representative actograms show data collected with motion detectors, and demonstrate activity bouts (indicated by arrows) resulting from introduction of novel running wheels in six spiny mice. No effects on phase of the rhythms were seen, even in two of the mice (shown here) which exceeded 2500 wheel revolutions (monitored independently) in the 3-h bout of activity that ensued upon introduction of the wheel to the cage. Lines superimposed on the actograms represent predicted phase of the activity rhythms based on regression of activity onsets for 7 days prior to introduction of novel wheel.

10 weeks. Comparisons of phase angle of entrainment, duration of daily activity, rates of re-entrainment to shifts in the light–dark cycle, and free-running period in constant darkness revealed no significant gender differences.

The temporal sensitivity of the circadian system to light exposure in spiny mice is similar to that seen for other species (e.g., hamsters, laboratory mice, rats), in which light exposure during the subjective daytime yields no significant changes in phase of circadian activity rhythms, while light exposure early in the subjective nighttime yields delays and light exposure in the latter portion of the subjective nighttime yields advances in the phase of rhythms. While differences in the magnitude of inducible phase advances versus phase delays for individual species are apparent in many previously published light PRCs [19–23], quantitative conclusions about these data should be drawn cautiously. The experimental protocol used may have introduced variability into the quantitative aspects of the measurements, considering that the data represent 3-h averages of individual responses of spiny mice simultaneously exposed to room lighting. While the spiny mice were observed to ensure wakefulness throughout the exposures, individuals may have received variable intensities of light throughout the 30-min exposure period due to shadows and positioning within their cages. Despite this variability, the shape of the PRC is shown to be similar to that of other nocturnal rodents, exhibiting the conservation of this phenomenon among mammalian species studied to date.

Free-running period is often used in the literature as a descriptor of the "intrinsic properties" of circadian rhythms in individual species, yet free-running period is a plastic property, as evident by the difference in free-running period of the two groups of mice observed in this study. Here, 15 of 16 spiny mice in Experimental Group 1 had free-running periods greater than 24.0 h, yet all of the mice ($n=6$) in Experimental Group 2 had free-running periods less than 24.0 h for the duration of the time in which they were monitored (2 months). Environmental or experimental factors have been shown to influence free-running period in rodents (e.g., light intensity [24]; wheel-running activity [25]). Such differences in the handling of these two groups of spiny mice, such as having been housed individually for several weeks longer, or having been subjected to two shifts of the light–dark cycle in the previous 2 months, may account for this difference.

Data from the re-entrainment experiments suggest a strong inhibitory effect of light on activity in spiny mice. The offsets of daily activity corresponded promptly with the onset of daily light even on the first days of the 10-day period of gradual re-entrainment of onsets to an advanced light–dark cycle. Direct inhibition of onsets of activity by light (i.e., "masking") during 6-h phase delays may have thus exaggerated measurements of rates of re-entrainment during delays. However, the coinciding rapid re-entrainment of both onsets and offsets of daily activity to a delayed LD cycle supports the position that spiny mice are more adept at achieving phase delays than phase advances.

Spiny mice showed no alteration in the phase of their locomotor rhythms in response to novel wheel-induced running activity. Large-amplitude shifting in the phase of activity rhythms in response to novel wheel-induced activity is an interesting phenomenon that may lend insight into biopsychological factors regulating the circadian system. While novel wheel shifts might have been observed in a larger sample of spiny mice or with introduction of a wheel at different times of the circadian cycle, it should be recognized that this phenomenon does not appear to be universal among mammals, having only been demonstrated previously in hamsters.

To summarize, the general properties of locomotor rhythms in adult spiny mice are generally similar to those of other species studied to date. They are nocturnal, with onset of activity coinciding promptly with onset of darkness. They are avid wheel runners, and access to running wheels yield significant changes to period and duration of daily activity, although novelty-induced wheel running appears to have no effect on phase of activity rhythms. Their circadian responses to light exposure are temporally similar to those of other species. Current studies are examining additional aspects of photic responsiveness, including range of entrainment and responses to changes in entraining photoperiod (long days, short days, skeleton photoperiods).

Spiny mice display rare characteristics for rodents [11] that may be useful for additional studies of the circadian system. For example, their propensity to maintain collective groups may provide a model to study interactions of social behavior and circadian rhythms, or their tendency to develop diabetes may offer insight into associations between metabolism and circadian rhythms. In conjunction with the knowledge of their adult circadian activity rhythms provided here, spiny mice may prove to be useful in studying interactions of the circadian system with other aspects of development, physiology and behavior in mammals.

Acknowledgements

The authors would like to acknowledge the technical support of Ms. Danielle Dunn and the kind gift of animals from Dr. Richard Deni (Rider University, Lawrenceville, PA, USA). This material is based on work supported by the National Science Foundation under Grant No. IBN-0235915.

References

[3] Kas MJ, Edgar DM. A nonphotic stimulus inverts the diurnal-nocturnal phase preference in *Octodon degus*. J Neurosci 1999;19:328–33.

[4] Blachong JA, McElhinny TL, Mahoney MM, Smale L. Nocturnal and diurnal rhythms in the unstriped nile rat, *Arvicanthis niloticus*. J Biol Rhythms 1999;14:364–77.
[5] Haim A, Rozenfeld FM. Temporal segregation in coexisting Acomys species: the role of odour. Physiol Behav 1993;54:1159–61.
[6] Friedman D, Haim A, Zisapel N. Temporal segregation in coexisting spiny mice (Acomys): role of photoperiod and heterospecific odor. Physiol Behav 1997;62:407–11.
[7] Gannon RL, Rea MA. Twelve-hour phase shifts of hamster circadian rhythms elicited by voluntary wheel running. J Biol Rhythms 1995;10:196–210.
[8] Rea MA, Barrera J, Glass JD, Gannon RL. Serotonergic potentiation of photic phase shifts of the circadian activity rhythm. NeuroReport 1995;6:1417–20.
[9] Turek FW, Losee-Olson S, Starz KE. Triazolam phase shifts the circadian clock of hamsters: implications for circadian abnormalities in humans. Clin Neuropharmacol 1986;9(Suppl. 4):83–5.
[10] Moriya T, Yoshinobu Y, Ikeda M, Yokota S, Akiyama M, Shibata S. Potentiating action of MKC-242, a selective 5-HT1A receptor agonist, on the photic entrainment of the circadian activity rhythm in hamsters. Br J Pharmacol 1998;125:1281–7.
[11] Brunjes PC. The precocial mouse, *Acomys cahirinus*. Psychobiology 1990;18:339–50.
[12] Weaver DR, Reppert SM. Maternal-fetal communication of circadian phase in a precocious rodent, the spiny mouse. Am J Physiol 1987; 253:E401–9.
[13] Weaver DR, Reppert SM. Direct in utero perception of light by the mammalian fetus. Dev Brain Res 1989;47:151–5.
[14] Weaver DR, Reppert SM. Definition of the development transition from dopaminergic to photic regulation of c-fos gene expression in the rat suprachiasmatic nucleus. Brain Res Mol Brain Res 1995;33: 136–48.
[15] Watkins JB, LaFollette JW, Sanders RA. Biotransformation in Egyptian spiny mouse *Acomys cahirinus*. Comp Biochem Physiol 1995;110C:101–7.
[16] Birke L, Sadler D. Maternal discrimination and the development of sex differences in exploratory behavior in infant spiny mice (*Acomys cahirinus*). Q J Exp Psychol 1991;43B:403–30.
[17] Rea MA, Michel AM, Lutton LM. Is Fos expression necessary and sufficient to mediate light-induced phase advances of the suprachiasmatic circadian oscillator? J Biol Rhythms 1993;8:S59–64.
[18] Takahashi JS, Menaker M. Interaction of estradiol and progesterone: effects on circadian locomotor rhythm of female golden hamsters. Am J Physiol 1980;239:R497–504.
[19] Daan S, Pittendrigh CS. A functional analysis of circadian pacemakers in nocturnal rodents: II. The variability of phase response curves. J Comp Physiol, A 1976;106:253–66.
[20] Takahashi JS, DeCoursey PJ, Bauman L. Spectral sensitivity of a novel photoreceptive system mediating entrainment of mammalian circadian rhythms. Nature 1984;308:186–8.
[21] Summer TL, Ferraro JS, McCormack CE. Phase-response and Aschoff illuminance curves for locomotor activity rhythm of the rat. Am J Physiol 1984;246:R299–304.
[22] Jiao YY, Lee TM, Rusak B. Photic responses of suprachiasmatic area neurons in diurnal degus (*Octodon degus*) and nocturnal rats (*Rattus norvegicus*). Brain Res 1999;817:93–103.
[23] Mahoney M, Bult A, Smale L. Phase response curve and light-induced fos expression in the suprachiasmatic nucleus and adjacent hypothalamus of *Arvicanthis niloticus*. J Biol Rhythms 2001;16:149–62.
[24] Aschoff J. Circadian rhythms: influences of internal and external factors on the period measured in constant conditions. Z Tierpsychol 1979;49:225–49.
[25] Weisgerber D, Redlin U, Mrosovsky N. Lengthening of circadian period in hamsters by novelty-induced wheel running. Physiol Behav 1997;62:759–65.